

## The mean

The mean is the sum of the values divided by their number.

$$\bar{x} = \frac{\sum x}{n}$$

**What is the mean of synthetic counts 2, 3, 5, 6?**

## **A mean with units**

Synthetic file sizes: 10, 15, 20, 35 KB.

**What is their mean, including units?**

## Recover a total

Five synthetic downloads have a mean duration of 8 seconds.

**What is their total duration?**

## Sample and population means

$\bar{x}$  denotes a sample mean;  $\mu$  denotes a population mean.

**Which symbol describes the mean size of all files on a server?**

## The median

Synthetic data. The median is the middle ordered value, or the average of the two middle values.

**What is the median of 9, 2, 7, 4, 3?**

## An even-sized sample

Synthetic downloads per session: 0, 1, 2, 2, 4, 9.

**What is the median?**

## The mode

Synthetic data. A mode is a value with greatest frequency.

**What is the mode of 0, 1, 2, 2, 2, 4, 4?**

## Two modes

Synthetic downloads per session: 0, 1, 1, 3, 3, 4.

**Which values are modes?**

## **A categorical mode**

Synthetic notebook colors: blue, black, blue, red, black, blue.

**Which measure of center is meaningful for these categories?**

## Compare mean and median

Synthetic file sizes (KB): 4, 5, 5, 6, 30.

**How do the mean and median compare?**

## Sensitivity to one value

Synthetic file sizes (KB): 4, 5, 5, 6, 30. Replace 30 with 300.

**Which measure of center changes more?**

## Choosing a typical value

Most files are small, but a few archive files are enormous.

**Which center would better describe a typical file?**

## A mean from frequencies

Synthetic print-job counts.

| Pages | Jobs |
|-------|------|
| 1     | 2    |
| 2     | 3    |
| 5     | 1    |

**What is the mean number of pages per job?**

## Why weight the values?

Synthetic job frequencies: one page 2 times, two pages 3 times, five pages 1 time.

**Why is  $(1 + 2 + 5)/3$  not the mean per job?**

## A grouped estimate

Synthetic file sizes are grouped into intervals.

| Size (KB) | Frequency |
|-----------|-----------|
| 0 to <10  | 3         |
| 10 to <20 | 5         |
| 20 to <30 | 2         |

**What midpoints would you use to estimate the mean?**

## Estimate a grouped mean

Synthetic file sizes: intervals  $[0,10)$ ,  $[10,20)$ ,  $[20,30)$  have frequencies 3, 5, 2.

**What mean results from weighting the class midpoints?**

## Exact or estimated?

A grouped table gives only interval counts, not individual values.

## Why is a midpoint-based mean generally an estimate?

## Combine two groups

Synthetic group A: 10 jobs, mean 4 pages.  
Group B: 30 jobs, mean 8 pages.

**What is the mean for all 40 jobs?**

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## Adding a constant

Every synthetic print job gains a two-page cover sheet.

**How does the mean pages per job change?**

## Changing units

Synthetic file sizes have mean 2 MB. Use 1 MB = 1,000 KB for this problem.

**What is the mean in KB?**